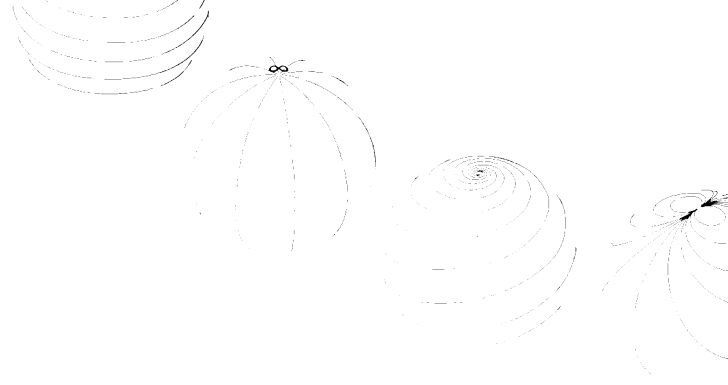




CHAPTER 10

Vector Fields: Physics and Topology

10.1 Vector Fields

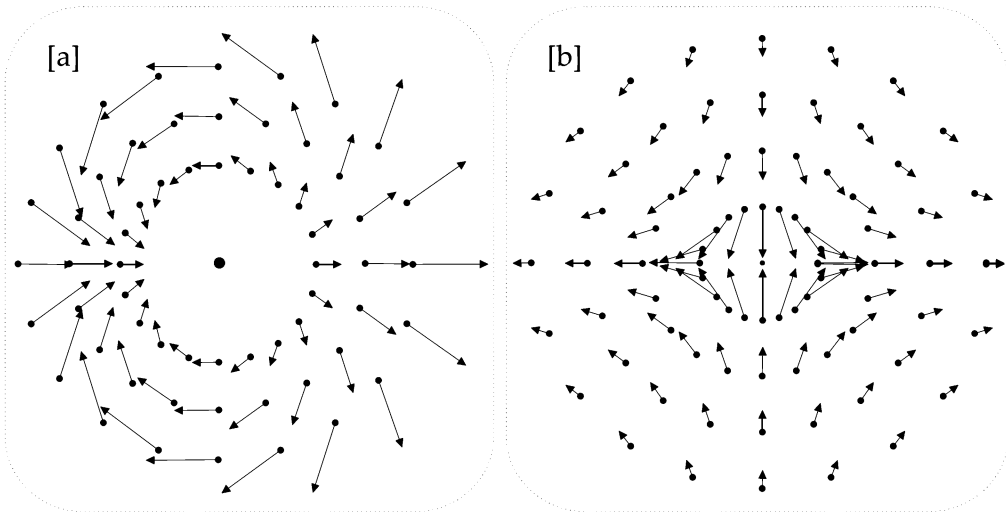
10.1.1 Complex Functions as Vector Fields

Throughout the course of this book we have relied on a *single* means of visualizing a complex function, namely, as a mapping of points in one complex plane to points in another. This idea has proved to be extremely powerful, for in terms of it the complex derivative is nothing more complicated than a local amplification. Despite its many virtues, in this chapter we shall abandon the mapping paradigm and introduce a completely new one in its place, thereby gaining a host of fresh insights into the subject and revealing surprising connections with physics.

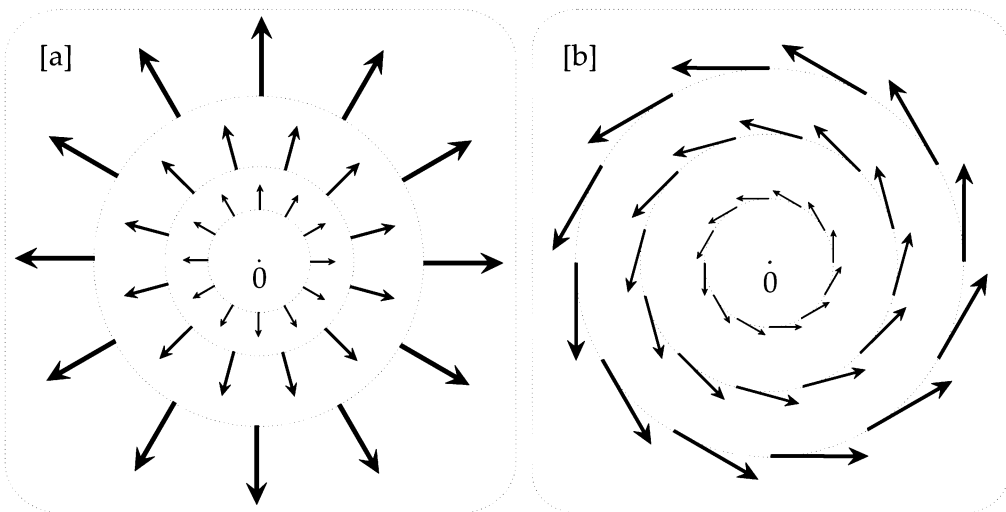
The new picture of a complex function $f(z)$ involves only a single complex plane. As before, the variable z is thought of as a point in this plane, but now comes the new idea: *the value of $f(z)$ is pictured as a vector emanating from z* . The resulting diagram of points with attached vectors is called the *vector field* of f . Figures [10.1a] and [10.1b] illustrate the vector fields of z^2 and $(1/z)$, respectively; before reading further you should study them carefully and convince yourself of their correctness. Try doing a sketch of the vector fields of some other powers, then compare them with accurate ones done by your computer. Also use the computer to examine the vector fields of e^z , $\log z$, and $\sin z$.

The vector field concept remedies a significant defect in the mapping point of view. Although we can learn a lot about a mapping by looking at the images of specific shapes, we do not get a feel for its overall behaviour. But if we let our eyes roam over the vector field of a complex function we *do* get such a view, in much the same way as we can survey the behaviour of a real function by scanning its graph.

Just as a complex mapping determines a vector field, so a vector field determines a mapping—the two concepts are equivalent. More explicitly, given the vector V issuing from the point z , we translate the tail of V to the origin and define the image of z to be the point at the tip.



[10.1] Vector field representation of $f(z)$: Draw the complex number $f(z)$ emanating from z . **[a]** The vector field of $f(z) = z^2$. **[b]** The vector field of $f(z) = (1/z)$.



[10.2] A vector field defines a mapping. **[a]** This vector field corresponds to the mapping $z \mapsto (1/2)z$. **[b]** This vector field corresponds to the mapping $z \mapsto (1/2)iz$.

Consider the examples in [10.2a] and [10.2b]. If z lies on a circle of radius r then the vector field in [10.2a] is radial and has length $(r/2)$; in [10.2b] the vector field has the same length but is tangential instead of radial. Check that when viewed as mappings, [10.2a] corresponds to an expansion of the plane by $(1/2)$, while [10.2b] corresponds to the same expansion followed by (or preceded by) a rotation through a right angle.

If the vectors in [10.2a] were instead directed inwards, what would the corresponding mapping be? If the vectors in [10.2b] were flowing clockwise, what would the corresponding mapping be?

10.1.2 Physical Vector Fields

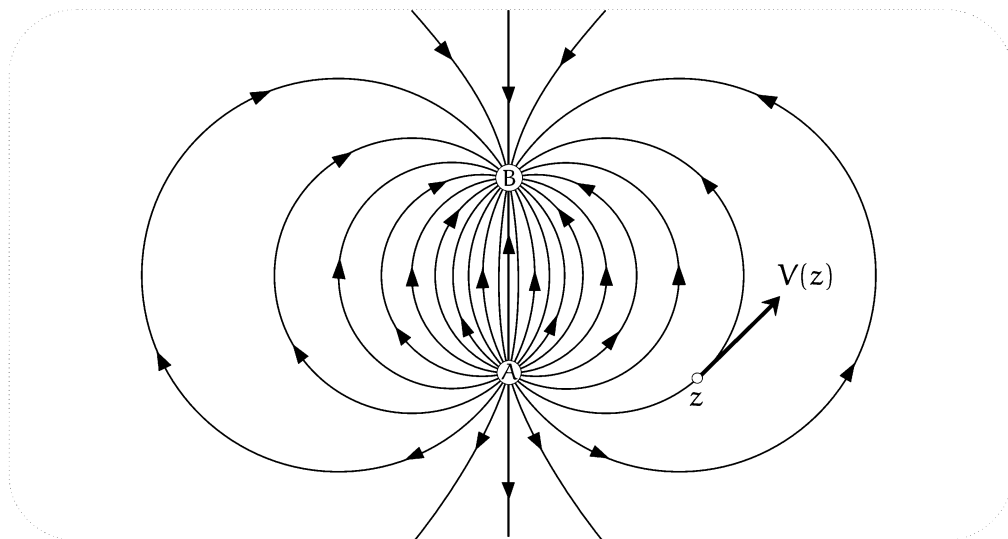
Since a vast range of physical phenomena find their most natural description as vector fields, the potential utility of the new way of looking at complex mappings should be obvious.

For example, consider the astonishingly complex array of electromagnetic disturbances zipping through the space around you. The visible light carrying these words to your retina, the totality of television and radio programs simultaneously being broadcast to your home—all this constitutes only a small part of the frenzied activity. But it is a remarkable fact that this great tangle of signals is in fact completely described by just two vector fields! At each instant of time t there is an electric vector $\mathbf{E}(p, t)$ and a magnetic vector $\mathbf{B}(p, t)$ emanating from each point p in space, and these two vector fields constitute the complete description of the electromagnetic field.

If we are to describe such physical vector fields with complex mappings, two problems immediately present themselves. A television set is fixed in space, and the way in which it produces its picture is by monitoring how the electromagnetic vector fields at its location *vary in time*. But a complex mapping is a timeless thing—it assigns the vector $f(z)$ to the point z *once and for all*. This is the first of our two problems. Thus if we are not to radically alter our conception of a complex mapping, the only types of physical vector fields we can describe in this manner are those that do *not* vary with time. We shall call such vector fields *steady*.

Fortunately, steady vector fields are both common and important in physics. For example, the unwavering character of the orbits of the planets reflects the fact that the gravitational field of the sun does not vary with time. In fact Newton informs us that this time-independent force on a particle of unit mass located at a point p in space may be represented by a vector emanating from p , directed to the centre c of the sun, and with a length equal to $M/[cp]^2$, where M is the mass of the sun. Drawing these vectors throughout space we obtain a steady vector field.

The above examples of the electromagnetic and gravitational vector fields illustrate our second problem—they exist in *three-dimensional* space, whereas the complex plane can only accommodate a *two-dimensional* vector field. There is no getting around this problem, but once again it is fortunate that there are certain important types of physical phenomena which are intrinsically two-dimensional in nature, and which can therefore be described in the complex plane. Let us begin with the flow of electricity within a sheet of conducting material.



[10.3] The Phase Portrait of $V(z)$ is the set of streamlines along which V flows: mathematically, they are the curves that have V as their tangent vectors. Given such a picture, one can immediately deduce the direction of $V(z)$ at z , but its magnitude is invisible. However, in many physical examples it is possible to draw the phase portrait in a special way, such that the closer together the streamlines, the stronger the flow. This is the case here, where electrodes are attached to a metal plate at A and B , and electricity flows between them—you see that the spacing is smallest, and therefore the flow greatest, along AB . Note that the streamlines are arcs of circles; this is proved geometrically in the next figure.

Take two wires and connect them to a battery, then touch the ends to two points A and B of a thin copper plate. Almost instantly a steady flow of electric current from one electrode to the other will be set up in the plate. See [10.3]. At each point z of the plate we now represent this flowing current by a time-independent vector in the direction of the flow, and with a length equal to the strength of the current there. Picturing the plate as a portion of $\mathbb{C}$, the flow is thus expressed as a complex function $V(z)$.

Rather than drawing the actual vector field in [10.3] we have instead shown the paths along which the electricity flows. Such a picture is called the *phase portrait* of the vector field, and the directed curves along which the flow occurs are called the *integral curves* or *streamlines* of the vector field. As illustrated, the streamlines of this example are in fact arcs of circles connecting the two electrodes. We shall justify this shortly.

Phase portraits are easy to take in visually and are thus a common way of representing vector fields. By definition the vector field is everywhere tangent to the streamlines, and thus its direction can be recovered from the phase portrait. On the other hand, it would seem that a phase portrait would necessarily fail to include the information about the *lengths* of the vectors. This is true in general, but for

many vector fields that arise in physics it will be shown that there exists a special way of drawing the phase portrait so that the strength of the flow is manifested as the crowding together of the streamlines: *the closer together the streamlines, the stronger the flow*¹. Later we shall explain this idea in detail, but for the moment we remark that [10.3] has actually been drawn in this special way. For example, as we approach the line-segment connecting the electrodes the streamlines become more and more crowded together, corresponding to a stronger and stronger current.

10.1.3 Flows and Force Fields

One and the same vector field (or phase portrait) can represent many quite different physical phenomena. For example, reconsider the copper plate in [10.3], and imagine that it is now sandwiched between two layers of material which do not conduct heat. Remove the electrodes and instead of supplying electricity at a constant rate at A, let us supply heat. Likewise, let us remove heat at the same constant rate at B. After a short time a steady pattern of heat flow from A to B will be established within the copper plate. In this steady state we may assign to each point a vector in the direction that the heat is flowing there, and having a length equal to the intensity of the heat flow.

Remarkably, in this steady state the physical laws governing the behaviour of the heat are identical to those which previously described the electricity, and thus the phase portrait [10.3] for the electric current is *also* the phase portrait for the new heat flow.

Here is yet another interpretation of [10.3]. In attempting to understand the flow of real liquids, such as water, it is helpful to consider an idealized fluid with the properties of being frictionless, incompressible, and “irrotational”—the precise meaning of the last term will be explained later. Imagine a thin layer of such an ideal fluid sandwiched between two horizontal plates, one of which has two small holes, A and B. If we now connect the holes with a fine tube that passes through a pump, a steady flow will be set up in the layer of fluid, and at each point we may draw its velocity vector. The phase portrait of this steady vector field is *once again* given by [10.3]!

Although there are certainly important differences between these three interpretations of [10.3], we may nevertheless lump them together in one class, for they are all flows of something. Whether it is electricity, heat, or liquid, in each case the vector field can be thought of as the velocity of flowing “stuff”, and the streamlines are the paths along which this stuff flows.

A physically quite distinct class is comprised of *force fields*. For example, although we previously discussed how the gravitational field of the sun could be represented as a steady vector field, the vector at a point in space is no longer the

¹ Faraday was the first to conceive of vector fields in this way; Maxwell then rendered the idea mathematically precise and exploited it to the hilt.

velocity of some flowing substance, rather, it represents the *force* experienced by a unit mass placed there. In the context of force fields, integral curves are called *lines of force* rather than streamlines. Here the lines of force are rays coming out of, or rather entering, the centre of the sun. Although this force field is three-dimensional, spherical symmetry² means that it will be the same on any plane drawn through the centre of the sun. It can therefore be completely described by a complex function.

Although there is nothing actually flowing along the lines of force, we can switch back to the flow point of view by *pretending* that there is, thereby interpreting a force field as the velocity field of a flowing substance. This is not mere sophistry: it is a remarkable fact that for the most common and important force fields (e.g., gravitational and electrostatic) *this imaginary flowing substance behaves exactly like our previously considered ideal fluid*.

To illustrate this, we turn to an example in electrostatics: equal and opposite charges (per unit length) are induced on two long wires which are then held parallel to each other in empty space. To each point in space we now attach the force vector that a unit electric charge would experience there; this force field is (by definition) the electric field E , and its phase portrait is the same on each plane drawn perpendicular to the wires. Taking [10.3] to be such a plane, with the wires piercing through at A and B , the phase portrait of this force field is exactly the one shown there for the flow of ideal fluid.

10.1.4 Sources and Sinks

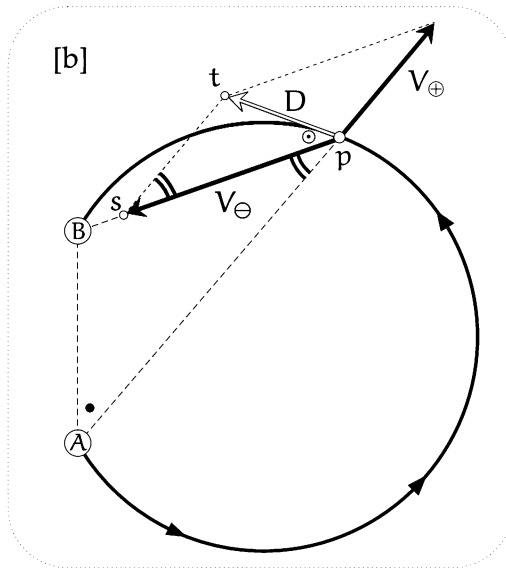
In order to make a quantitative analysis of [10.3], we introduce the concepts of (two-dimensional) sources and sinks. Thinking in terms of our layer of ideal fluid, a *source of strength* S is a point at which we pump in S units of fluid per unit of time. Figure [10.4a] illustrates the symmetric velocity vector field $V(z)$ of an isolated source at the origin.

Given a curve (open or closed) in a general flow, the amount of fluid flowing across it in each unit of time is called the *flux*. Clearly the flux across an element of the curve is just its length times the component of the velocity perpendicular to the curve. The total flux across the curve is then the sum (i.e., integral) of these elementary fluxes. Returning to the specific case of [10.4a], our assumption of incompressibility says that the flux across any simple loop round 0 must be the same as the amount of fluid S being pumped in at 0. Since the flow is orthogonal to the origin-centred circle C of radius r , we deduce that

$$2\pi r |V| = S.$$

Writing $z = r e^{i\theta}$, we find that the vector field of the source is therefore

² This is an idealization—like the earth, the sun is somewhat flattened at the poles.


$$V(z) = |V| e^{i\theta} = \frac{S}{2\pi} \left(\frac{e^{i\theta}}{r} \right) = \frac{S}{2\pi} \left(\frac{1}{\bar{z}} \right).$$
$$V_{\oplus}(z) = \frac{S}{2\pi} \left(\frac{1}{\bar{z} - \bar{A}} \right).$$
$$V_{\ominus}(z) = -\frac{S}{2\pi} \left(\frac{1}{\bar{z} - \bar{B}} \right).$$

We now know the vector fields $V_{\oplus}(z)$ and $V_{\ominus}(z)$ which would be produced by the source or the sink in [10.3] if each were present on its own, but what is the flow when they are both present together? [Incidentally, this combination of a source

and a sink of equal strength is called a *doublet*.] The answer is perhaps slightly clearer if we switch to the equivalent electrostatic problem of parallel charged wires through A and B. A unit charge at z is repelled by A with force $V_{\oplus}(z)$, and attracted by B with force $V_{\ominus}(z)$. The *net* force $D(z)$ of the doublet acting on the charge is then simply the *vector sum* of the two separate forces:

$$D(z) = V_{\oplus}(z) + V_{\ominus}(z) = \frac{S}{2\pi} \left(\frac{1}{\bar{z} - \bar{A}} - \frac{1}{\bar{z} - \bar{B}} \right) = \frac{S}{2\pi} \frac{(\bar{A} - \bar{B})}{(\bar{z} - \bar{A})(\bar{z} - \bar{B})}. \quad (10.1)$$

We will now show geometrically that, as claimed in [10.3], the net force at p is tangent to the circle through A, p , and B. Consider [10.4b]. It is easy to see [exercise] that D will be tangent to the circle if and only if the angles marked $\bullet$ and $\odot$ are equal, so this is what we must demonstrate. As illustrated, the angles ApB and pst are clearly equal. But we also have

$$\frac{ts}{ps} = \frac{|V_{\oplus}|}{|V_{\ominus}|} = \frac{Bp}{Ap}.$$

Thus the two shaded triangles are similar, and therefore $\bullet = \odot$, as was to be shown.

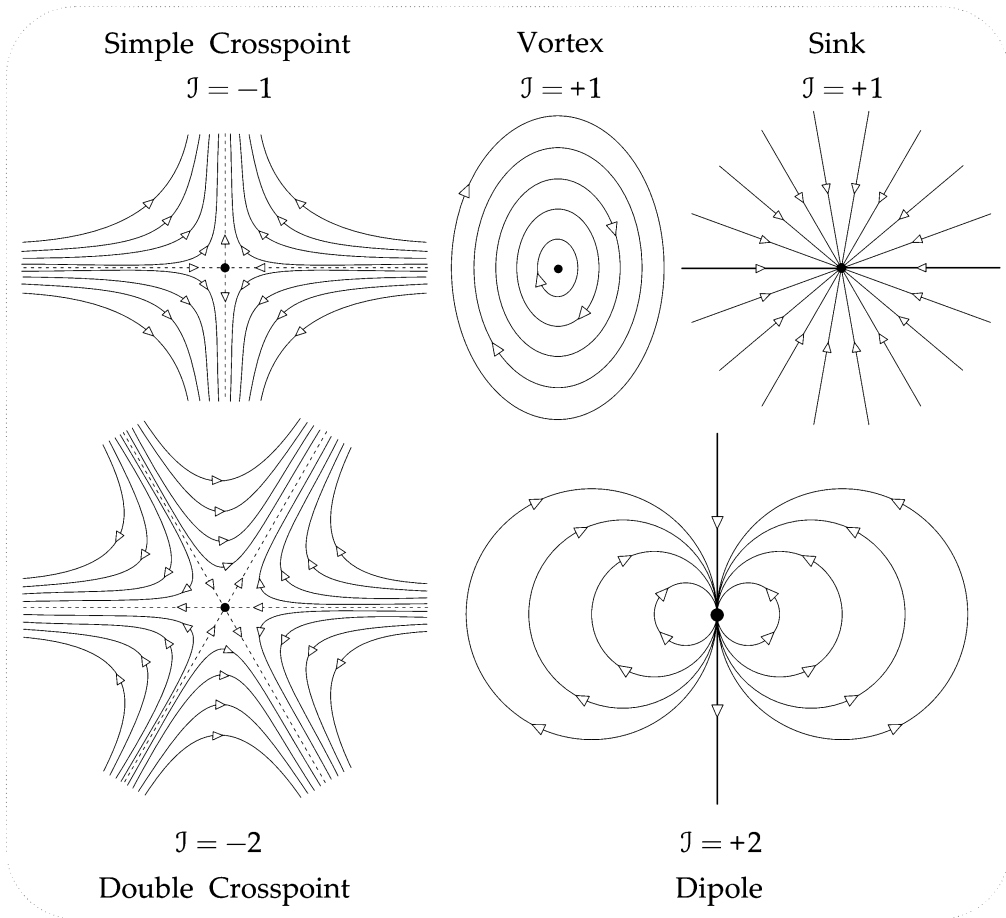
10.2 Winding Numbers and Vector Fields*

10.2.1 The Index of a Singular Point

Let us confine all our discussions to vector fields for which the direction is well-defined and continuous at all but a finite number of points. These exceptional places, where the vector field vanishes or becomes infinite, are called *singular points*³. They are easy to spot in a phase portrait, usually as the intersection points of distinct streamlines. Figure [10.5] shows the phase portraits in the vicinity of some simple types of singular points, together with their names and their “indices”—a term which we must now explain.

Figure [10.6] is a magnified view of the simple crosspoint (also called a *saddle point*) shown in [10.5]. Round this singular point s we have drawn a simple loop Γ_s , and at some of its points we have also drawn the vectors V . Since Γ_s does not pass through any singular points, the direction of V is well-defined and continuous everywhere on it. Thus we can count the net number of revolutions of $V(z)$ as z traverses Γ_s . We call this number the *index* $\mathcal{I}_V[\Gamma_s]$ of the loop Γ_s with respect to the vector field V . When it is clear which vector field is being considered, we may simplify this notation to $\mathcal{I}[\Gamma_s]$. For example, in [10.6] we see that $\mathcal{I}[\Gamma_s] = -1$. Note that we have drawn the vectors on Γ_s only to make it easier to see the value of the index; actually, since only the *directions* of the vectors are required, the phase portrait is sufficient on its own.

³ Otherwise known as *critical points* or *singularities*—terms to which we have already attached different meanings.

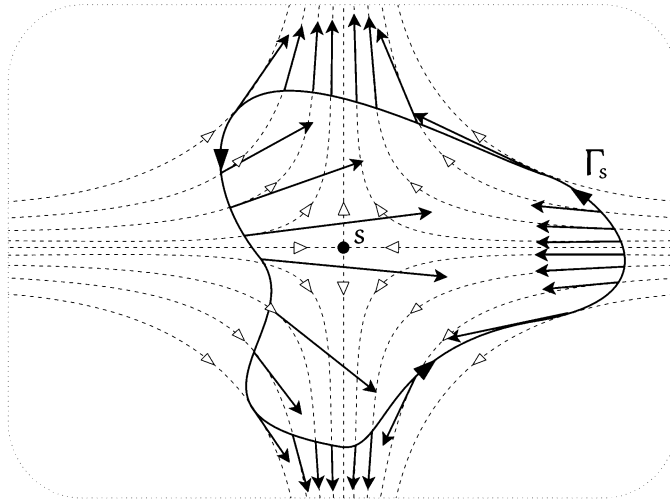


[10.5] The values of the index (J) for various singular points. The geometric definition of J is explained in the next figure.

If we continuously deform Γ_s without crossing s (or any other singular point) then the value of $J[\Gamma_s]$ will also vary continuously, and since it's an integer, it will therefore remain constant. Thus we may unambiguously define the *index of a singular point* s to be the index of any loop that winds round s once, but does not wind round any other singular points. It should not cause any confusion if we abuse our notation slightly and call this index $J(s)$. Applying this definition to loops of your choosing, you may now verify each of the given values of J in [10.5].

Before moving on, we observe three properties of the index:

- (i) There is nothing to stop us applying the above definition to a *non-singular* point, but in this case the index must vanish. Choosing Γ_s to be a very small loop, the non-singular nature of s implies that all the vectors on Γ_s will point in roughly the same direction, and so $J(s) = 0$.



[10.6] The index of a singular point is the net number of revolutions executed by the vector field as we traverse a simple (counterclockwise) loop around the singular point. Here, $J = -1$.

- (ii) If V undergoes a certain rotation as we traverse a piece of curve, then $(-V)$ undergoes the same rotation. Thus if we reverse the direction of the flow in each of the above phase portraits, the index will remain the same. For example, a source must have the same index as a sink, namely, $J = 1$.
- (iii) Just as the index is insensitive to the precise shape of Γ_s , so it is insensitive to the precise shape of the streamlines. Imagine that [10.6] is drawn on a rubber sheet which we gradually stretch, so producing a new distorted phase portrait. The direction of V at each point of Γ_s will undergo a continuous change, and so its net revolutions upon traversing Γ_s will likewise vary continuously. The index must therefore remain constant.

Clearly, our new concept of “index” is related to our old concept of “winding number”, but how? If we instead think of V as a mapping, sending the points of Γ_s to those of a new loop $V(\Gamma_s)$, then a moment of thought reveals that the index of Γ_s is just a new interpretation of the winding number of its image loop:

$$J_V[\Gamma_s] = \nu[V(\Gamma_s), 0]. \quad (10.2)$$

This makes it clear [see p. 397] that the index $J(s)$ of a point s is the same thing as its *topological multiplicity* $\nu(s)$ as a preimage of 0. In particular, if V is analytic then

$$J(\text{root of order } n) = n \quad \text{and} \quad J(\text{pole of order } m) = -m.$$

Check this for the examples in [10.1].

If you have not done so yet, we urge you to use a computer to draw the vector fields of some simple polynomials and rational functions. Notice how roots and poles show up just as vividly as the corresponding x -intercepts and vertical

asymptotes occurring in the graph of a real function. Notice how easy it is to zoom in on the vector field to find their precise locations. But now we can do much better than the real case, for we can also immediately read off the *precise nature* of a root or singularity simply looking at its index!

The following example illustrates the fact that a vector field contains *more* information than an ordinary graph. If we sketch the graphs of

$$F(x) = \frac{(x-1)^2}{(x+2)^3} \quad \text{and} \quad G(x) = \frac{(x-1)^4}{(x+2)^7}$$

the results will be qualitatively the same: both look something like a parabola near $x = 1$; both have branches going to opposite ends of the vertical asymptote at $x = -2$; both look something like $(1/x)$ for large x .

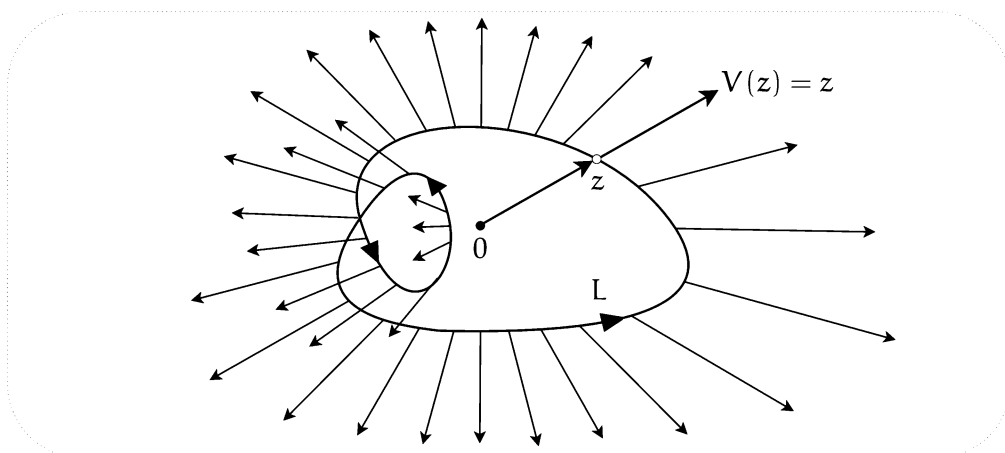
Now use the computer to draw the corresponding vector fields when x is replaced by z . Striking indeed are the differences! As we traverse a small loop around the root at $z = 1$, F makes two positive revolutions while G makes four; doing the same at the pole $z = -2$, F makes three negative revolutions while G makes seven; and on a very large origin-centred circle, F makes one negative revolution while G makes three.

Returning to the general significance of (10.2), consider the ordinary winding number $v[L, 0]$ of a loop L . This can now be viewed as the index of L with respect to the vector field of the identity mapping:

$$v[L, 0] = J_z[L].$$

Figure [10.7] illustrates this result with $J_z[L] = 1$. The winding number of L around a general point a is likewise just its index with respect to the vector field $(z - a)$:

$$v[L, a] = J_{(z-a)}[L].$$



[10.7] Index of the vector field $V(z) = z$ is the winding number of the loop: $v[L, 0] = J_z[L]$.

10.2.2 The Index According to Poincaré

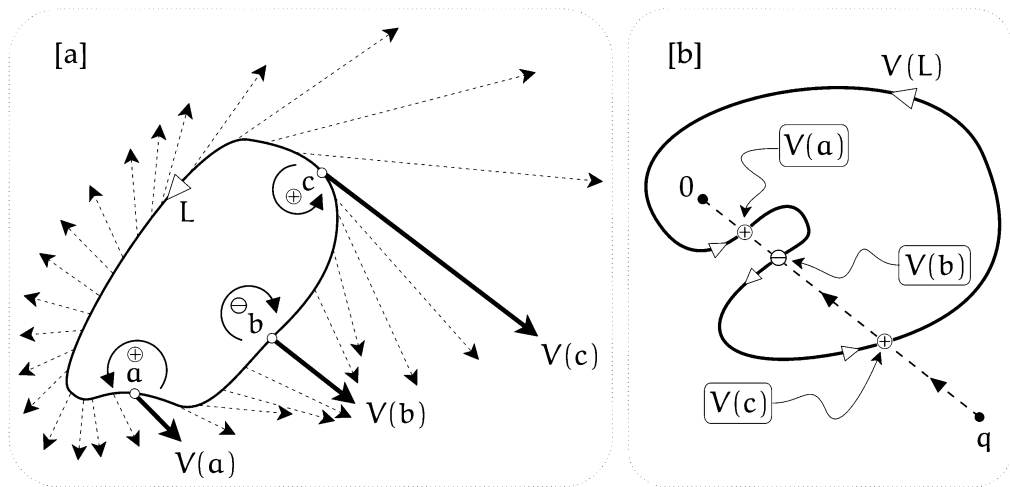
Figure [10.8a] shows a loop L and a vector field V evaluated on it. Let us use this simple example (for which it is obvious that $J_V[L] = 1$) to explain a quick method (due to Poincaré) of finding the index in more complicated cases.

Consider all the places on L (a, b, c in our case) where V points in one arbitrarily chosen direction. Let P be the number of these places at which $V(z)$ rotates in the positive sense as z passes through it, and let N be the number at which it rotates in the negative sense. Even in relatively complicated cases, P and N are usually quick and easy to find. We now obtain the index as the difference of these two numbers:

$$J_V[L] = P - N. \quad (10.3)$$

In our case $P = 2$ because of the positive rotation at a and c , and $N = 1$ because of the negative rotation at b . Thus $J_V[L] = 1$, as it should.

To see just how fast and efficient (10.3) is to use, try it out on each of the examples in [10.5], choosing L to be a counterclockwise circle centred at s , and choosing the direction to be vertically upward. For extra practice, try it again with a different choice of direction.



[10.8] Poincaré's method for quickly finding the index. [a] First, choose an arbitrary direction, then find all the places z_j on L where $V(z_j)$ points in this direction. Count the number P of z_j 's where $V(z)$ is rotating in the positive, counterclockwise direction as we pass through z_j , and likewise count the number N of places where it is rotating in the negative direction. Then, $J_V[L] = P - N$. In the illustrated case, $P = 2$, $N = 1$, and so the formula correctly predicts that $J_V[L] = 2 - 1 = 1$. [b] Consider the image $V(L)$ of L under the mapping corresponding to the vector field V . On the ray in the chosen direction, start far far away at q , where the winding number $\nu = 0$. The intersection points of the ray with $V(L)$ are the images $V(z_j)$ of the z_j 's, and a positive rotation of V at z_j corresponds to a left-to-right crossing, so the crossing rule (7.1) increases ν by one. Likewise, a negative rotation of V reduces ν by one. Thus we have proved Poincaré's formula.

Although the truth of (10.3) is probably clear at an intuitive level, it is nevertheless instructive to deduce it from the “crossing rule” (7.1), p. 388, for computing winding numbers.

Figure [10.8b] shows the image $V(L)$ of L when V is viewed as a mapping. [Check that it really *is* the image!] In these terms, the required index is just $\nu[V(L), 0]$. Draw the ray from 0 in the previously chosen direction, and let the point q travel along it (starting far away), ending up at the origin. In its journey, q will thus cross $V(L)$ at the points $V(c)$, $V(b)$, and $V(a)$. In the vector field picture [10.8a], the positive rotation of V at c now implies (in [10.8b]) that q sees $V(L)$ directed from left to right as it approaches the first crossing at $V(c)$. Oppositely, the negative rotation at b implies that $V(L)$ is directed from right to left as q approaches $V(b)$. But as we previously argued in Chapter 7, $\nu[V(L), 0]$ is the number of points P at which $V(L)$ is directed from left to right (as seen by q as it approaches), minus the number of points N at which it is directed from right to left. Done.

10.2.3 The Index Theorem

With the connection between indices and winding numbers established, the Topological Argument Principle can be reinterpreted in terms of vector fields:

The index of a simple loop is the sum of the indices of the singular points it contains.

Using a neater argument than the one given in Chapter 7, we can now extend this theorem to *multiply connected* regions. As illustrated in [10.9], recall that this means that the region has holes in it; two in our case.

The shaded region consists of the points which are inside C and outside B_1 and B_2 . In general there could be more holes, say g of them, with counterclockwise boundary curves $B_1, B_2, \dots, B_g$. As illustrated, suppose that we have a vector field on such a region, and let $s_1, s_2, \dots, s_n$ be the singular points within the region. In our case there are only two: s_1 is a dipole, and s_2 is a saddle point. The generalization of the Argument Principle is this:

$$\mathcal{J}[C] - \sum_{j=1}^g \mathcal{J}[B_j] = \sum_{j=1}^n \mathcal{J}[s_j].$$

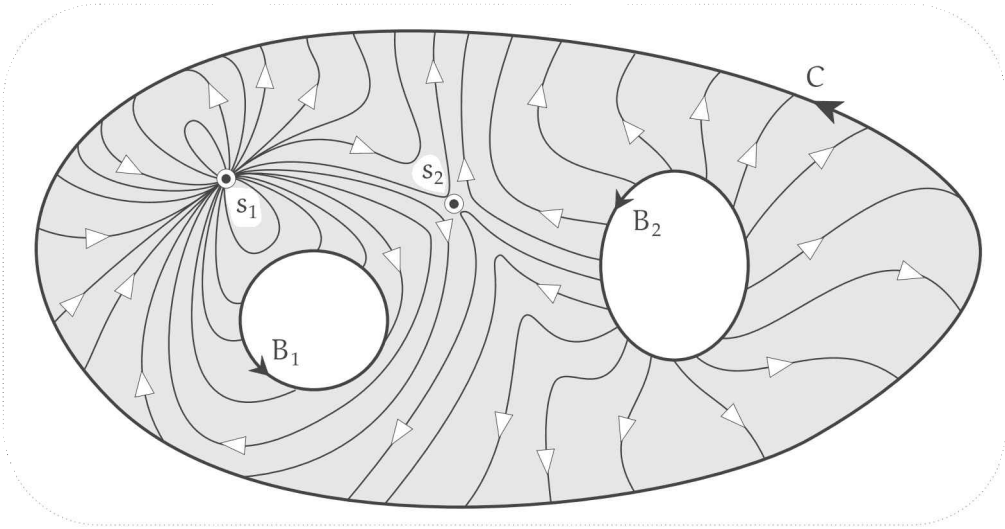
It is called the *Index Theorem*.

Perhaps using (10.3), for practice, verify that in our example, $\mathcal{J}[C] = 2$, $\mathcal{J}[B_1] = 0$, and $\mathcal{J}[B_2] = 1$, so that the LHS of the Index Theorem equals 1. But the RHS is

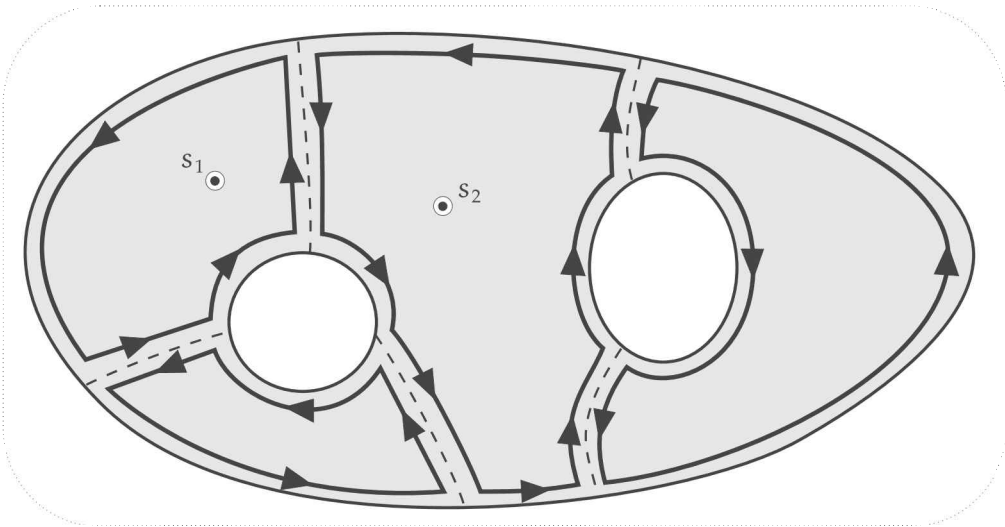
$$\mathcal{J}(\text{dipole}) + \mathcal{J}(\text{saddle point}) = 2 + (-1) = 1,$$

so confirming the prediction of the theorem in this case.

To understand this result, consider [10.10]. Using the dashed curves, break the region into curvilinear polygons in such a way that each one contains at most one



[10.9] The Index Theorem: $\mathcal{I}[C] - \sum^g \mathcal{I}[B_j] = \sum^n \mathcal{I}[s_j]$. This is certainly true in the illustrated example, for $\mathcal{I}[C] - \mathcal{I}[B_1] - \mathcal{I}[B_2] = 2 - 0 - 1 = 1$, and $\mathcal{I}[s_1] + \mathcal{I}[s_2] = 2 + (-1) = 1$.



[10.10] Proof of the Index Theorem. Using the dashed curves, break the region into curvilinear polygons in such a way that each one contains at most one singular point, and let their counterclockwise boundaries be K_j . If we sum the indices of all the K_j 's then we obtain the RHS of the Index Theorem. On the other hand, the index of a single K_j is obtained by looking at how much the vector field rotates as one travels along each edge of K_j , then adding up these net rotation angles. But when we sum the indices of all the K_j 's, each interior edge [dashed] is traversed twice, once in each direction, and the associated angles of rotation therefore cancel. The remaining edges of the K_j 's together make up C and $-B_1, -B_2$, etc. Summing the associated angles of rotation (divided by 2π) therefore yields the LHS of the Index Theorem. Done.

singular point, and let their counterclockwise boundaries be K_j . If we sum the indices of all the K_j 's then we obtain the RHS of the Index Theorem. For if K_j does not contain a singular point then its index vanishes, while if it does contain one then its index is (by definition) the index of that singular point.

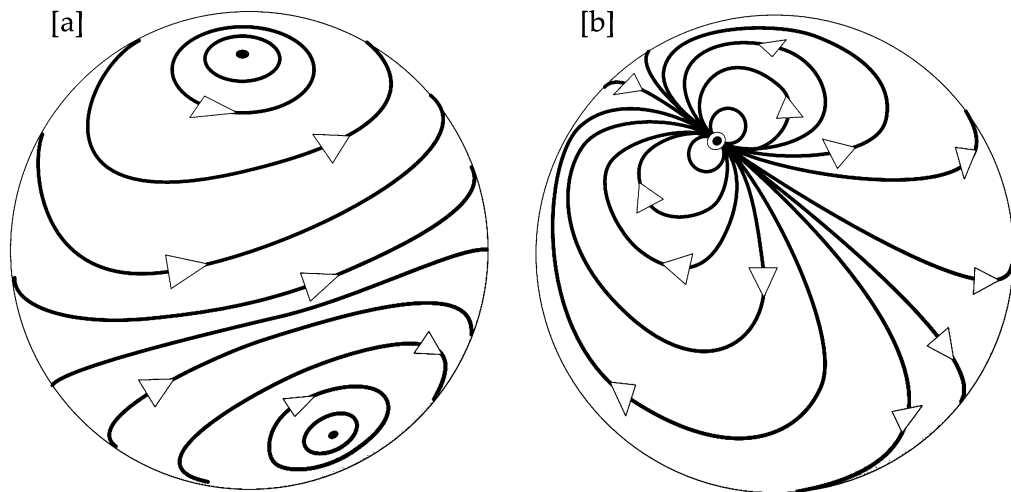
On the other hand, the index of a single K_j is obtained by looking at how much the vector field rotates as one travels along each edge of K_j , then adding up these net rotation angles. But when we sum the indices of all the K_j 's, each interior edge [dashed] is traversed twice, once in each direction, and the associated angles of rotation therefore *cancel*. The remaining edges of the K_j 's together make up C and $-B_1, -B_2$, etc. Summing the associated angles of rotation (divided by 2π) therefore yields the LHS of the Index Theorem. Done.

10.3 Flows on Closed Surfaces*

10.3.1 Formulation of the Poincaré–Hopf Theorem

If a curved surface S in space is “smooth” in the sense that there exists a tangent plane at each of its points, then it makes sense to speak of a vector field that is everywhere tangent to S . Intuitively, we may picture such a vector field as the velocity of a fluid that is flowing over S .

Figure [10.11] shows the streamlines of two such flows on the sphere. Notice that both possess singular points: [10.11a] has two vortices, while [10.11b] has a dipole. In fact there can be *no* vector field on the sphere that is free of singular points. This is



[10.11] Poincaré–Hopf Theorem: If a vector field on a smooth surface of genus g has only a finite number of singular points, then the sum of their indices is $\chi = (2 - 2g)$. Two examples: [a] $\mathcal{I}(\text{vortex}) + \mathcal{I}(\text{vortex}) = 1 + 1 = 2 = \chi(\text{sphere})$. [b] $\mathcal{I}(\text{dipole}) = 2 = \chi(\text{sphere})$.

one consequence of an extremely beautiful result called the *Poincaré–Hopf Theorem*, the formulation of which we will now sketch.

It is not immediately obvious how to give a precise definition of the “index” of a singular point on a curved surface, but for the moment let us accept that this integer exists, and that its value is the same as for an analogous singular point in the plane. Thus if we sum all the indices in [10.11a] we obtain

$$\mathcal{J}(\text{vortex}) + \mathcal{J}(\text{vortex}) = 1 + 1 = 2,$$

while if we do the same for [10.11b] we obtain

$$\mathcal{J}(\text{dipole}) = 2.$$

Try drawing your own streamlines on an orange, then sum the indices of the singular points. Is this a coincidence??

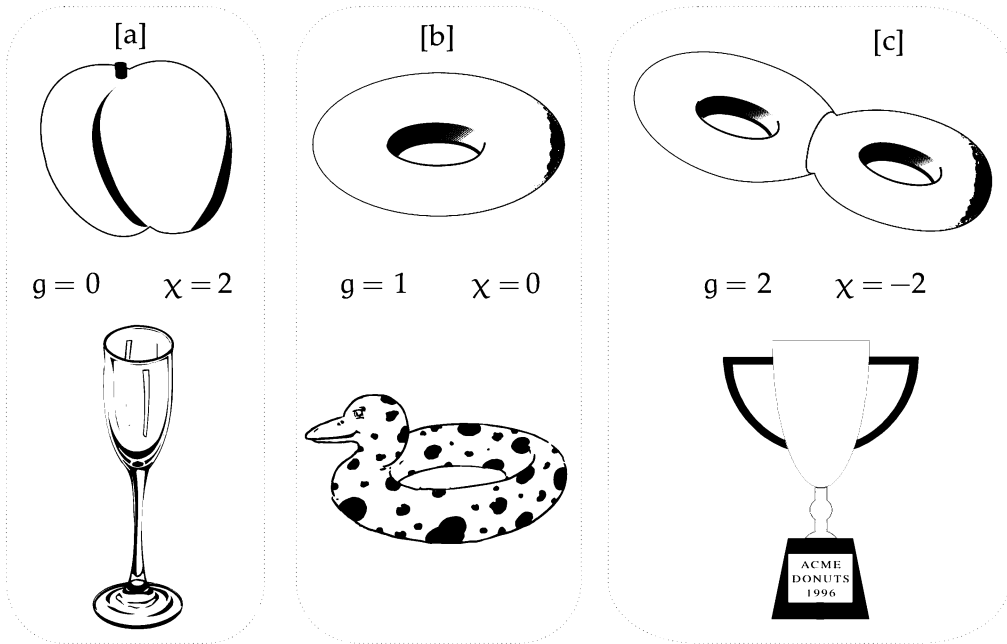
There *are* no coincidences in mathematics! In the case of the sphere, the Poincaré–Hopf Theorem states that if we sum the indices of *any* vector field on its surface, we will always get 2 for the answer. Indeed, it says that we will get this answer for any surface that is *topologically* a sphere, that is to say, any surface into which the sphere may be changed by a continuous and invertible transformation. If we imagine the sphere to be made of rubber, examples of such transformations and surfaces are given by stretching without tearing. The surfaces of the plum and the wineglass in [10.12a] are two examples of such topological spheres.

The sphere is the boundary of a solid ball, and other closed surfaces may likewise be obtained as the boundaries of other solid objects. For example, the surface of a doughnut is called a torus (top of [10.12b]), and it is clear that this surface is topologically the same as the beach toy at the bottom. But it seems equally clear that no amount of stretching and bending can turn these surfaces into a sphere—[10.12a] and [10.12b] are topologically distinct types of surface. Figure [10.12c] shows yet a third topologically distinct class. Obviously we could continue this list indefinitely just by adding more holes.

We shall not develop the topological ideas⁴ necessary to prove it, but once again it seems clear that these classes of topologically distinct closed surfaces can be classified purely on the basis of their number of holes. This number g is called the *genus* of the surface (see [10.12]). We can now formulate the general result:

$$\text{If a vector field on a smooth surface of genus } g \text{ has only a finite number of singular points, then the sum of their indices is } \chi \equiv (2 - 2g). \quad (10.4)$$

⁴ See “Further Reading”, at the end of this chapter.



[10.12] The genus (g) and the Euler characteristic (χ) of a closed surface. The genus counts the number of holes in a closed surface. The Euler characteristic $\chi \equiv (2 - 2g)$ turns out to be ubiquitous in topology, and it is therefore a more natural way of classifying surfaces. Furthermore, χ in fact has its own, independent geometrical definition, and it is then a theorem that $\chi = (2 - 2g)$. See Ex. 12.

The number $\chi \equiv (2 - 2g)$ occurring in this theorem is called the *Euler characteristic* of the surface, and it turns out to be ubiquitous in topology. It is therefore more natural to classify our surfaces using χ rather than g . See [10.12]. Furthermore, χ in fact has its own, independent geometrical definition, and it is then a *theorem* that $\chi = (2 - 2g)$. See Ex. 12.

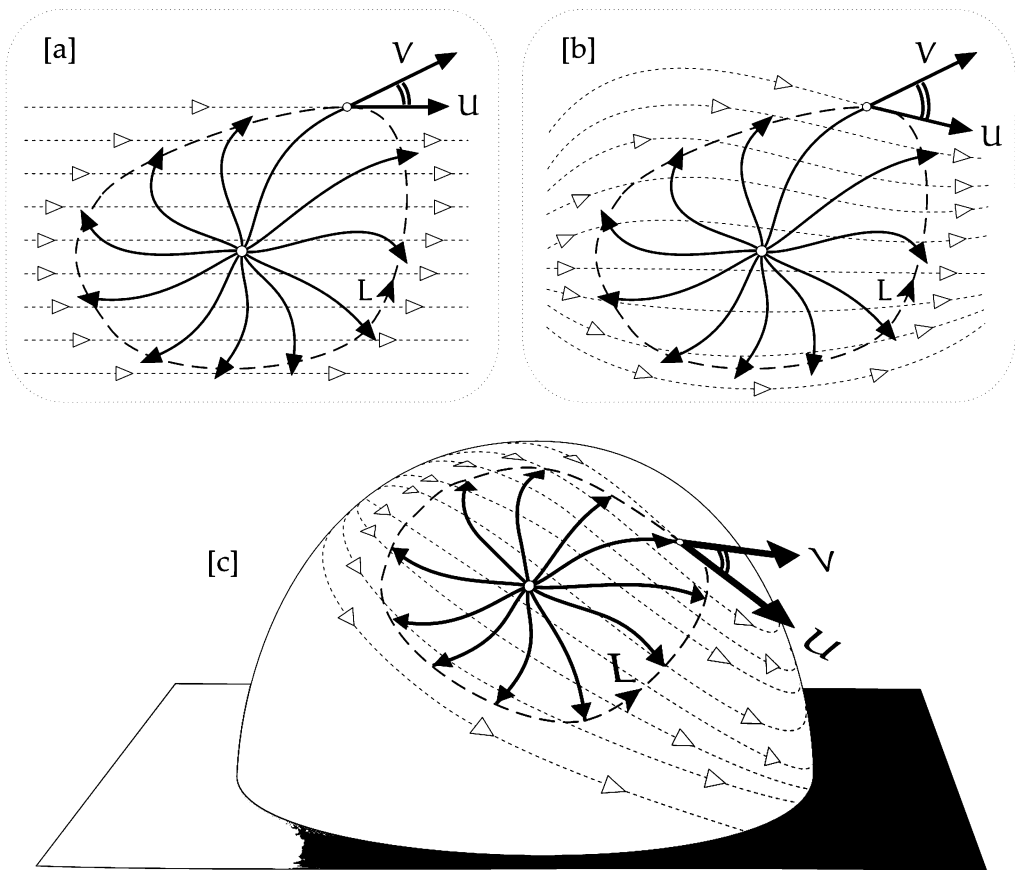
An immediate consequence of (10.4) is that a vector field without any singular points can exist only on surfaces of vanishing Euler characteristic, i.e., the topological doughnuts. Even then, the theorem does not actually guarantee that such a vector field exists, it merely says that if there are singular points then their indices must cancel. However, you can see for yourself [draw it!] that on a doughnut there *do* exist at least two topologically distinct vector fields without any singular points: one that circulates around its axis of symmetry, and one that circulates through the hole.

10.3.2 Defining the Index on a Surface

In order to give a precise definition of the “index” of one of the singular points in [10.11], we should presumably draw a loop round it on the surface, then find the

net rotation of the vector field as the loop is traversed. But wait, *rotation relative to what?*

To answer this question, we first re-examine the familiar concept of rotation in the plane. Figure [10.13a] shows that (in the plane) the rotation of $V(z)$ along L can be thought of as taking place relative to a *fiducial vector field* having horizontal streamlines, say $U(z) = 1$. If we define $\angle UV$ to be the angle between U and V , and let $\delta_L(\angle UV)$ be the net change in this angle along L , then our old definition of the index is



[10.13] Definition of the index of a singular point on a curved surface. [a] The original definition of the index in the plane can be thought of as the net rotation of V relative to a horizontal fiducial vector field U : $\mathcal{I}_V[L] = \frac{1}{2\pi} \delta_L(\angle UV)$. [b] The definition and value of the index does not change if we replace U with any vector field that is nonsingular on and inside L . [c] Now imagine that [b] is drawn on a rubber sheet. If we continuously stretch it into the form of the curved surface then not only will $\frac{1}{2\pi} \delta_L(\angle UV)$ remain well-defined, but its value will not change.

$$\mathcal{I}_V [L] = \frac{1}{2\pi} \delta_L (\angle UV). \quad (10.5)$$

If we continuously deform the horizontal streamlines of U in [10.13a] to produce those in [10.13b] then, by the usual reasoning, the RHS of (10.5) will not change. Thus we conclude that this formula yields the correct value of the index if we replace U with *any* vector field that is nonsingular on and inside L .

Now imagine that [10.13b] is drawn on a rubber sheet. If we continuously stretch it into the form of the curved surface in [10.13c] then not only will the RHS of (10.5) remain well-defined, but its value will not change. To summarize: if s is a singular point of a vector field V on a surface S , we define its index as follows. Draw any nonsingular vector field U on a patch of S that covers s but no other singular points; on this patch, draw a simple loop L going round s ; finally, apply (10.5), that is count the net revolutions of V relative to U as we traverse L .

10.3.3 An Explanation of the Poincaré–Hopf Theorem

We can now give a very elegant derivation of theorem (10.4), due to Hopf (1956) himself. The argument proceeds in two steps. First, we show that on a surface of given genus, all vector fields yield the same value for the sum of their indices; second, we produce a concrete example of a vector field for which the sum equals the Euler characteristic. This proves the result.

Suppose that V and W are two different vector fields on a given closed surface S . See [10.14]. If v_j are the singular points of V (marked $\bullet$) and w_j are those of W (marked $\odot$), we must show that

$$\sum \mathcal{I}_V [v_j] = \sum \mathcal{I}_W [w_j].$$

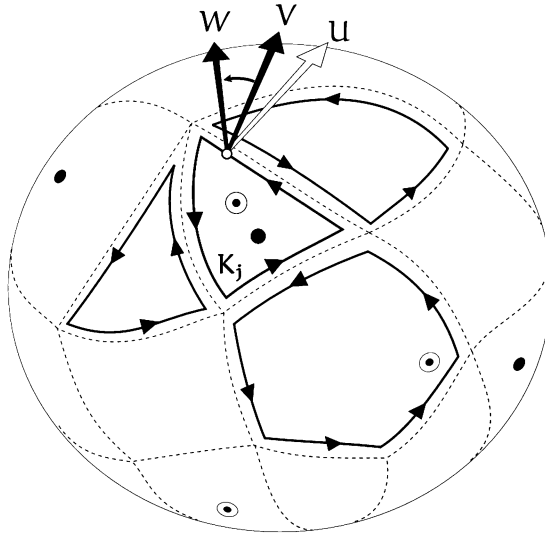
Much as we did in [10.10], we divide up S into curvilinear polygons (dashed) such that each one contains at most one v_j and one w_j , taken counterclockwise as viewed from outside S . To find the indices of V and W along boundary of one of the polygons K_j , draw any nonsingular vector field U on the polygon and then use (10.5). The difference of these indices is then

$$\begin{aligned} \mathcal{I}_W [K_j] - \mathcal{I}_V [K_j] &= \frac{1}{2\pi} [\delta_{K_j} (\angle UW) - \delta_{K_j} (\angle UV)] \\ &= \frac{1}{2\pi} \delta_{K_j} (\angle VW), \end{aligned}$$

which is explicitly independent of the local vector field U .

From this we deduce that

$$\begin{aligned} \sum \mathcal{I}_V [v_j] - \sum \mathcal{I}_W [w_j] &= \sum_{\text{all polygons}} (\mathcal{I}_V [K_j] - \mathcal{I}_W [K_j]) \\ &= \frac{1}{2\pi} \sum_{\text{all polygons}} \delta_{K_j} (\angle VW) \\ &= 0, \end{aligned}$$

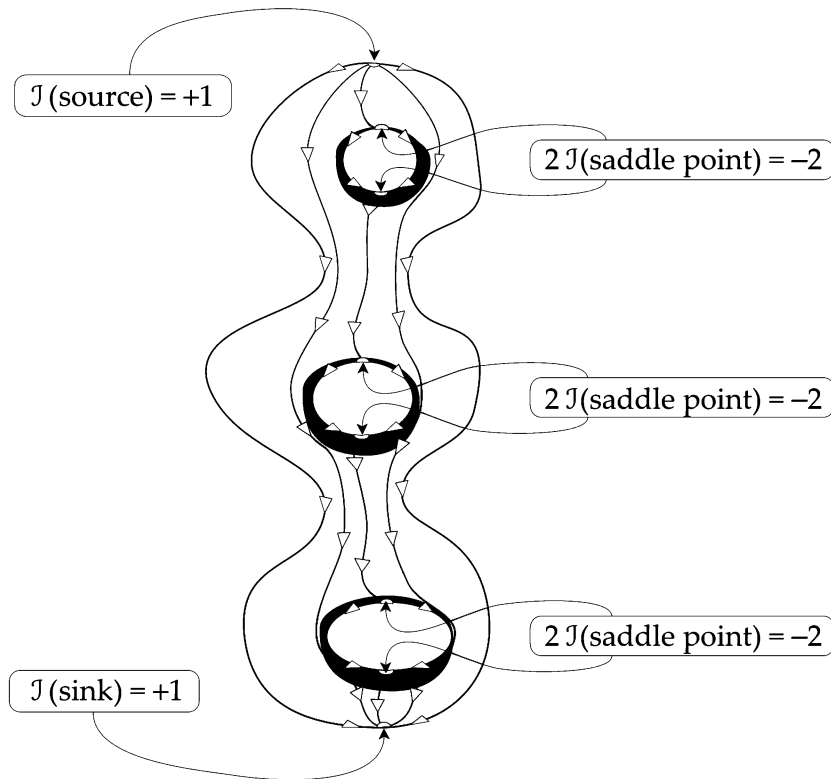


[10.14] Proof that the sum of the indices is the same for all vector fields. Let V (with singular points v_j marked $\bullet$) and W (with singular points w_j marked $\odot$) be two different vector fields on the surface S . Divide up S into curvilinear polygons (dashed) such that each one contains at most one v_j and one w_j . Now consider one of these polygons and its boundary K_j . Draw any nonsingular vector field U on the polygon. Then $\mathcal{I}_W[K_j] - \mathcal{I}_V[K_j] = \frac{1}{2\pi} [\delta_{K_j}(\angle UW) - \delta_{K_j}(\angle UV)] = \frac{1}{2\pi} \delta_{K_j}(\angle VW)$, which is explicitly independent of the local vector field U . If we now sum this over all polygons, each edge is traced twice, in opposite directions, so $\sum \delta_{K_j}(\angle VW) = 0$, and therefore $\sum \mathcal{I}_V[K_j] = \sum \mathcal{I}_W[K_j]$. Done.

because every edge of every polygon is traversed once in each direction, producing equal and opposite changes in $\angle VW$. We have thus completed the first step: the sum of the indices is independent of the vector field.

Since the index sum for the example in [10.11a] is 2, we now know that this is the value for any vector field on a topological sphere. The second step of the general argument is likewise to produce an example on a surface of arbitrary genus g , such that the sum is $\chi = (2 - 2g)$. Figure [10.15] is such an example for $g = 3$, the generalization to higher genus being obvious. Here we imagine that honey is being poured onto the surface at the top—it then flows over the surface, finally streaming off at the very bottom. As the figure explains, and as was required, the sum of the indices is indeed equal to χ .

Further Reading. These topological ideas—in combination with ideas in the next two chapters—open the door to the important subject of *Riemann surfaces*. In particular, we hope you will find it easier to read [Klein \(1881\)](#), which champions



[10.15] A specific vector field on a surface of genus g for which the sum of the indices is χ . We imagine that honey is being poured onto the surface at the top—it then flows over the surface, finally streaming off at the very bottom. As the figure explains, and as was required, the sum of the indices is indeed χ . But we know that this sum is the same for all vector fields on the surface, and we have therefore shown that the sum of their indices must always equal χ , completing the proof of the Poincaré–Hopf Theorem.

Riemann’s original approach to multifunctions in terms of fluid flowing over a surface in space. See also [Springer \(1981, Ch. 1\)](#), which essentially reproduces Klein’s 1881 monograph, but with additional helpful commentary. For a good introduction to the more abstract, modern view of Riemann surfaces, see [Jones and Singerman \(1987\)](#). Finally, for more on topology itself, we recommend [Hopf \(1956\)](#), [Prasolov \(1995\)](#), [Stillwell \(2010\)](#), [Fulton \(1995\)](#), and [Earl \(2019\)](#).

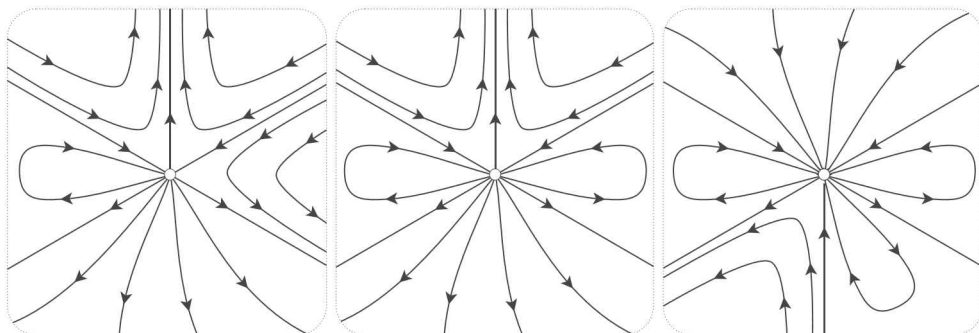
10.4 Exercises

- 1 Show both algebraically and geometrically that the streamlines of the vector field z^2 are circles that are tangent to the real axis at the origin. Explain why the same must be true of the vector field $1/\bar{z}^2$.
- 2 Use a computer to draw the vector field of $1/(z \sin^2 z)$. Use this picture to determine the location and order of each pole.
- 3 Use a computer to draw the vector field of

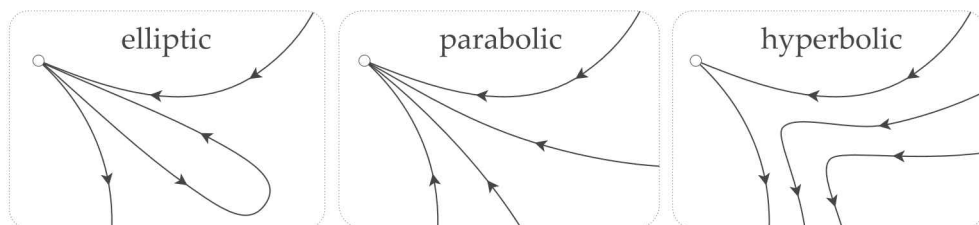
$$P(z) = z^3 + (-1 + 5i)z^2 + (-9 - 2i)z + (1 - 7i).$$

Use this picture to factorize $P(z)$, and check your answer by multiplying out the brackets.

- 4 Suppose that one of the streamlines of a vector field V is a simple closed loop L . Explain why L must contain a singular point of V .
- 5 Find the index of each of the three singular points shown below.



- 6 Observe that the neighbourhood of every singular point we have examined in this chapter is made up of sectors of one of the three types shown below, called *elliptic*, *parabolic*, and *hyperbolic*. Let e , p , and h denote the number of each type of sector surrounding a singular point.

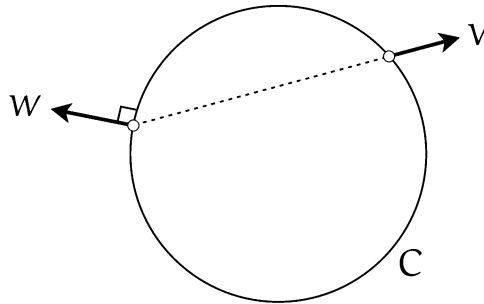


- (i) Verify that the index of each of the three singular points in the previous question is correctly (and painlessly) predicted by *Bendixson's Formula*:

$$J = 1 + \frac{1}{2}(e - h).$$

- (ii) Explain this formula.

- 7 Given a vector field V , defined on a circle C , let a vector field W be constructed on C in the manner illustrated below. If $J_V[C] = n$, find $J_W[C]$. [This problem is taken from [Prasolov \(1995, Ch. 6\)](#), and the answer may be found there, too.]



- 8 If f and g are continuous, one-to-one mappings of the sphere S to itself, then their composition $f \circ g$ is too. Let us prove that *at least one of these three mappings must possess a fixed point*. We proceed by the method of contradiction.

- (i) Show if the result is false then, for each point p of S , the points p , $f(p)$, and $[f \circ g](p)$ must be distinct.
- (ii) In this case, deduce that there is a unique, directed circle C_p on S passing through these three points in the stated order.
- (iii) Imagine a particle orbiting on C_p at unit speed, and let $V(p)$ be its velocity vector as it passes through p . Since p was arbitrary, V is a vector field on S .
- (iv) By appealing to the Poincaré–Hopf Theorem, obtain the desired contradiction.

- 9 Continuing from the previous exercise, apply the result as follows:

- (i) By taking $g = f$, deduce that $(f \circ f)$ has a fixed point.
- (ii) By taking $g = (\text{antipodal mapping})$, deduce that either f has a fixed point, or f maps some point to its antipodal point.

- 10 Arbitrarily choose a collection of points $s_1, s_2, \dots, s_n$ on a closed, smooth surface S . By attempting to draw examples on the surface of a suitable fruit or vegetable, investigate the following claim: There exists a flow on S whose only singular points are $s_1, s_2, \dots, s_n$, and the type of singular behaviour (dipole, vortex, etc.) at all but one of these points may be chosen arbitrarily.

- 11 Imagine the surface of the unit sphere divided up into F polygons, the edges all being “straight lines on the sphere”, i.e., great circles. Let E and V be the total number of edges and vertices that result from dividing up the sphere in this way.

(i) Let $\mathcal{P}_n$ be an n -gon on the unit sphere. Use (6.9), p. 317, to show that

$$\mathcal{A}(\mathcal{P}_n) = [\text{angle sum of } \mathcal{P}_n] - (n - 2)\pi.$$

[Hint: Join the vertices of $\mathcal{P}_n$ to a point in its interior, thereby dividing it into n triangles.]

(ii) By summing over all polygons, deduce that

$$F - E + V = 2.$$

[This argument is due to Legendre (1794); the result itself is a special case of the result in the following exercise.]

- 12 Let S be a smooth closed surface of genus g , so that its Euler characteristic is $\chi(S) = 2 - 2g$. As in [10.14], let us divide S into F polygons, and let E and V be the total number of edges and vertices, respectively.

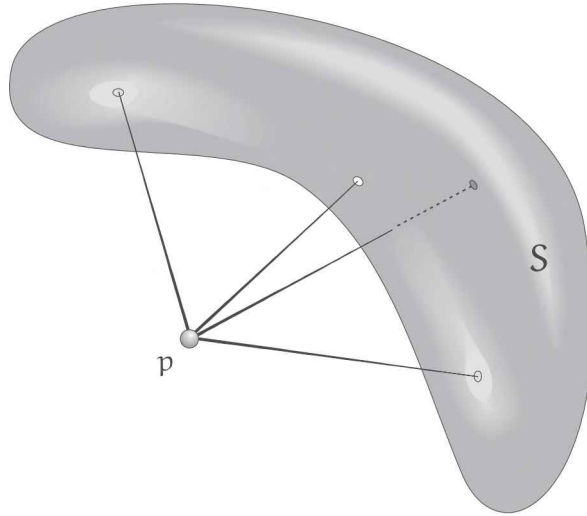
- (i) Draw a simple example on the surface of an orange and convince yourself (by drawing it) that we may obtain a consistent flow over the entire surface whose only singular points are (1) a source inside each of the F polygons; (2) a simple saddle point on each of the E edges; (3) a sink at each of the V vertices. We call this a *Stiefel vector field*, in honour of Hopf’s student, Eduard Stiefel [1909–1978]. See Frankel (2012, §16.2b). For an example of a Stiefel vector field, see the front cover of VDGF, Needham (2021).
- (ii) By applying the Poincaré–Hopf Theorem to such a flow on the general surface S , deduce the following remarkable result, which we call the *Euler–Lhuillier Formula*:

$$F - E + V = \chi(S).$$

NOTES: (1) This is actually the normal *definition* of χ , and it is then a *theorem* that $\chi(S) = 2 - 2g$. (2) Euler discovered the original version of this formula in 1750 for topological spheres, with $\chi = 2$; this is called *Euler’s Polyhedral Formula*. For a masterful, mathematically accurate, yet riveting account of this history and the connected mathematical ideas, see Richeson (2008).

(iii) Verify this result for your example in (i), then try it out on a doughnut.

- 13 The figure below shows all the normals that may drawn from the point p to the smooth surface S . Let $R(q)$ be the distance from p to a point q of S , and let us say that q is a *critical point* of R if the rate of change of R vanishes as q begins to move within S ; we need not specify the direction in which q begins to move because we are assuming that S has a tangent plane at q .



- (i) Explain why pq is normal to S if and only if q is a critical point of R .
- (ii) The level curves of R on S are the intersections of S with the “onion” of concentric spheres centred at p . Sketch these level curves in the vicinity of the illustrated critical points of R . Notice the distinction between points where R has a local maximum or minimum, versus points where R may increase or decrease depending on the direction (within S) in which one moves away from the critical point.
- (iii) Imagine that p generates an attractive force field, so that every point particle in space experiences a force F directed towards p . For example, we could imagine that p is the centre of the Earth, and that F is the Earth’s gravitational field. If a particle is constrained to move on S , then the only part of F to which it can respond is the projection F_S of F onto S . Sketch the streamlines of F_S . How are they related to the level curves of R in (ii)?
- (iv) You have just seen that the critical points of R are the singular points of F_S . How does the index $\mathcal{J}(q)$ of a singular point of F_S distinguish between the types of critical point discussed in (ii)?
- (v) Let us define the *multiplicity* of a normal pq to be this index $\mathcal{J}(q)$. Use the Poincaré–Hopf Theorem to deduce that

The total number of normals (counted with their multiplicities) that may be drawn to S from any point p is independent of both the location of p and the precise shape of S , and is equal to $\chi(S)$.

[This lovely result is essentially due to [Reech \(1858\)](#), though he did not express it in terms of $\chi(S)$, nor did he use an argument like the one above. With hindsight, Reech's work is a clear harbinger of Morse Theory, which it predates by some 70 years.]

- (vi) Verify Reech's theorem for a couple of positions of p in the case where S is a torus (doughnut).